## Core Modules

- Core modules cover topics that we feel every web developer should have a good grounding in
- This includes all the information they need to design and build a basic, accessible web app that follows modern best practices, and manage and deploy their code using appropriate tools
- Module list –
    1. Web standards
    2. Semantic HTML
    3. CSS fundamentals
    4. CSS text styling
    5. CSS layout
    6. JavaScript fundamentals
    7. Accessibility
    8. Design for developers
    9. Version control

## Web Standards

- This module covers the fundamentals of how the web works at a high level – including the model used for communication, the core technologies involved, how those technologies are created, and how a web browser renders and displays web sites to a user
- General resources (Resilient Web Design – Jeremy Keith)
- Topics are below –
    1. How the web works
    2. The HTML, CSS, and JavaScript triangle
    3. The web standards model
    4. How browsers load webpages

## Semantic HTML

- HTML is the technology that defines the content and structure of any website.
- Written properly, it should also define the semantics (meaning) of the content in a machine-readable way, which is vital for accessibility, SEO, and making use for built-in features browsers provide for content to work optimally
- This module covers the basics of the language, before looking at key areas such as document structure, links, lists, images, forms, and more
- General resources
    1. Structuring the web with HTML
    2. Learn HTML and CSS – Scrimba
- Topics are below –
    1. Basic HTML syntax
    2. Good document structure
    3. Lists
    4. Advanced text techniques
    5. Links
    6. Media
    7. Other interactive elements
    8. HTML tables
    9. Debugging HTML

## CSS fundamentals

- It enables you to add style to web pages including color, text, positioning and layout, animation.
- In this module, we cover all the fundamental language mechanics you need to understand how CSS works
- General resources (Learn HTML and CSS – Scrimba)
- Topics are below –
    1. Basic CSS syntax
    2. Selectors
    3. The box model
    4. Handling conflicts in CSS
    5. Values and units
    6. Sizing
    7. Backgrounds and borders
    8. Overflow
    9. Styling form elements
    10. Debugging CSS

## CSS text styling

- This module focuses specifically on CSS font and text styling, including loading custom web fonts and applying them to your text
- Topics are below –
    1. Text and font styling
    2. Styling lists and links
    3. Web fonts

## CSS layout

- Our final core CSS module looks deep into another crucial topic – creating layouts for modern websites.
- This module looks at floats, positioning, other modern layout tools, and building responsive designs that will adapt to different devices, screen sizes, and resolutions
- Topics are below –
    1. CSS layout basics
    2. Floats
    3. Positioning
    4. Modern layout
    5. Responsive design

## JavaScript fundamentals

- JavaScript is a huge topics, with different features, styles, and techniques to learn
- And so many APIs and tools built on top of it
- This module focuses mostly on the essentials of the core language, plus some key surrounding topics – learning these topics will give you a solid basis to work from
- General resources
    1. Learn JavaScript (Scrimba)
- Topics are below –
    1. Variables
    2. Math
    3. Text
    4. Arrays
    5. Conditionals
    6. Loops
    7. Functions
    8. JavaScript object basics
    9. DOM scripting
    10. Events
    11. Async JavaScript basics
    12. Network requests with fetch()
    13. Working with JSON
    14. Libraries and frameworks
    15. Debugging JavaScript

## Design for Developers

- The idea of this module is to (re-)introduce developers to design thinking
- They may not want to work as designers, but having some basic user experience and design theory is good for everyone involved in building websites, no matter what their role
- At the very last, even the most technical, non-designer developer should understand design briefs, why things are designed as they are, and be able to get into the mindset of the user
- And it'll help them make their portfolios look better
- - In addition, front-end developers are often tasked with doing various bits of design work on projects
- - Clients and employers often assume that they can do it because they are involved with the visual elements
- - Historically, web developer used to be more of a hybrid designer/developer role than it is today
- General resources –
    1. The Shape of Design (Frank Chimero)
    2. Designing for the Web (Mark Boulton)
    3. Design trampoline: Learn design theory basics (Anna Riazhskikh)
    4. Practical Typography (Matthew Butterick)
    5. Web typography made simple (Hanna Boom)
    6. Web Style Guide (Patrick J. Lynch and Sarah Horton)
    7. Visual design rules you can safely follow every time (Anthony Hobday)
    8. 16 little UI design rules that make a big impact (Adham Dannaway)
    9. Learn UI Design Fundamentals (Scrimba)
- Topics are below –
    1. Basic design theory
    2. User-centered design
    3. Design briefs

## Accessibility

- Access to web content such as public services, education, e-commerce sites, and entertainment is a human right
- No one should be executed based on disability, race, geography, or other human characteristics
- This module discusses the best practices and techniques should learn to make websites as accessible as possible
- Topics are below –
    1. Accessibility basics
    2. Accessible styling
    3. Accessible JavaScript
    4. Assistive technology
    5. WAI-ARIA

## Version Control

- Version control tools are an essential part of modern workflows, for backing up and collaborating on codebases
- This module takes you through the essentials of version control using Git and GitHub
- Learning outcomes –
    1. Understand why version control systems are necessary
        - Notes – Git and associated social coding sites like GitHub have a lot of functionality, and can be intimidating and unfriendly to begin with. In this module discuss the basic, and why it is necessary to have some experience here before entering the industry
        - Note – Git is the web industry standard for version control, and has been for some time
    2. Understand the difference between Git, and websites like GitHub and GitLab
    3. Understand GitHub to enable teamwork and collaboration that isn't so easy just with plain Git
    4. Basic setup – installing git, signing up for an account for your chosen social coding site
    5. Handling security requirements, like SSH/GPG keys
    6. Creating a repo
    7. Pushing changes – add, commit, and push
    8. Contribution to others' repos – forking | creating a new branch | creating a PR | review flow
    9. Using GitHub pages to publish a sample project
    10. Good housekeeping
        - Regularly update local repos to sync with their remote counterparts. This includes pulling remote changes to local repo, and installing package updates (with npm or yarn). Always do this before working on local repo
        - Use (.gitignore) to ignore all the stuff you don't want to commit. Examples include dependencies, dev source files, and OS-level admin files like (.DS_Store)
        - Delete branches you have finished with
    11. Handling merge conflicts